Daniel W. Cranston

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Website and publications · GitHub: graph-coloring-python

Summary

Computer scientist and mathematician with a PhD in computer science and a master's degree in applied mathematics focused on optimization and algorithms. Sixteen years of university research and teaching experience, specializing in graph theory, algorithm design, and discrete mathematics. Experienced in developing mathematical methods and algorithms, proving their correctness, and using computational experiments to investigate their behavior. Author of approximately 90 research papers and the graduate textbook *Graph Coloring Methods*. Strong record of communicating complex technical ideas through research publications, lectures, and presentations.

Technical Strengths

Algorithm design: Graph algorithms, combinatorial optimization, structural graph theory, algorithm correctness and mathematical proofs.
Computational experiments: Python, NetworkX, algorithm implementation, correctness checks, runtime measurement.
Research and communication: Research publications, technical writing, graduate-level instruction, presentations, collaboration.

Selected Computational and Algorithmic Research

Research topics: Graph/list/correspondence coloring, edge coloring, nowhere-zero flows, competitive optimization, recoloring and reconfiguration.
Methods: Kempe swaps, greedy algorithms, discharging and reducibility, the potential method.
Computational experiments: Implemented graph algorithms in Python using NetworkX; tested correctness, measured runtime, and evaluated performance across many graph instances.

Professional Experience

Virginia Commonwealth University — Richmond, Virginia 2009–2025
Associate Professor of Computer Science (2020–2025); *Associate Professor of Mathematics* (2015–2020); *Assistant Professor of Mathematics* (2009–2015)

- Conducted research in graph theory, algorithms, combinatorics, and discrete optimization.
- Published approximately 90 research papers, collaborating with researchers at institutions across the United States and internationally.
- Designed algorithms for graph-coloring problems and proved rigorous results about their correctness and applicability.
- Taught courses in algorithms, theory of computation, graph theory, combinatorics, multivariable calculus, linear algebra, introduction to proof, and related subjects.
- Mentored student projects that won awards within the department and the college of engineering.
- Communicated technical material to audiences of widely varying sizes and mathematical and computing backgrounds.

Johns Hopkins University 2016–2017
Visiting Associate Professor of Applied Mathematics and Statistics

DIMACS / Bell Labs 2007–2009
Postdoctoral Fellow — research in discrete mathematics and algorithms.

Education

University of Illinois Urbana-Champaign 1999–2007
Ph.D. in Computer Science, 2007; M.S. in Computer Science, 2004;
M.S. in Applied Mathematics (Optimization and Algorithms), 2001.
Greenville College 1996–1999
B.A. in Mathematics, magna cum laude, 1999.

Selected Publications and Technical Writing

- *Graph Coloring Methods* (2024). Graduate textbook, 453 pages. Freely available online.
- Nicolas Bousquet and Daniel W. Cranston, "A Linear Kernel for Independent Set Reconfiguration in Planar Graphs." *SIAM Journal on Discrete Mathematics*, to appear.
- Stijn Cambie, Wouter Cames van Batenburg, Daniel W. Cranston, Jan van den Heuvel, and Ross J. Kang, "Reconfiguration of List Colourings." Manuscript under review.
- Daniel W. Cranston and Reem Mahmoud, "Kempe Equivalent List-Colorings." *Combinatorica* 44 (2024), 125–153.
- Daniel W. Cranston, "List-Recoloring of Sparse Graphs." *European Journal of Combinatorics* 105 (2022), 103562.
- Daniel W. Cranston and Douglas B. West, "An Introduction to the Discharging Method via Graph Coloring." *Discrete Mathematics* 340 (2017), 766–793.
- Daniel W. Cranston and Candace M. Kent, "On the Boundedness of Positive Solutions of the Difference Equation $x_n = \max\left\{\frac{A^1_{n-1}}{x_{n-1}}, \frac{A^2_{n-1}}{x_{n-2}}, \ldots, \frac{A^t_{n-1}}{x_{n-t}}\right\}$ with Periodic Parameters." *Applied Mathematics and Computation* 221 (2013), 144–151.

Full publication list: arXiv publication page.

Professional Service

- Editor, *Electronic Journal of Combinatorics* (2025–present).
- Editorial Board, *Journal of Graph Theory* (2018–present) and *Discussiones Mathematicae Graph Theory* (2019–present).
- Managing Editor, *Australasian Journal of Combinatorics* (2018–2025).
- Guest Editor (with H.A. Kierstead and B. Rabern), special issue of *Discrete Mathematics* in honour of Landon Rabern, Volume 346, Issue 11 (November 2023).
- Refereed more than 130 research articles.